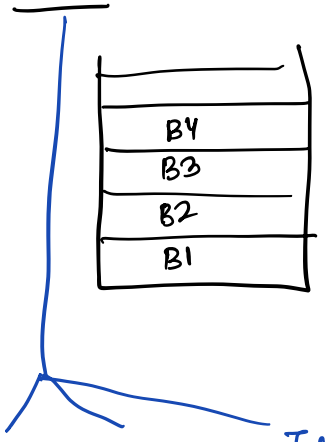
## Stacks

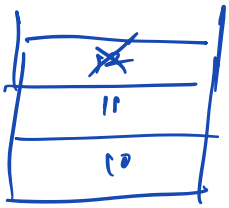



Entry and Exit are same

Stack is abstract datatype.

Push (Insert) — Pop (Delete) — Top. (Access top element of stack)

Stack s → s.push(12) → only at top.




s.pop() → 12 removed

top() → 11

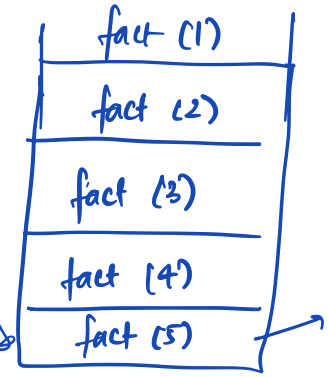
Size → how many elements in stack?

isEmpty → Boolean.

## Recursion Stack

5×4

fact(5)
↓
fact(4)
↓
fact(3)
↓
fact(2)
↓
fact(1) → fact(0)




fact(5) waiting for fact(4)

STACK Memory.

## Stack Using Array

Stack

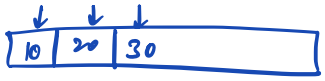
Array

Linked List

top

len(array) - 1

size




append of an array.

remove last element of array. pop()

---

Stack

private arr

push — append

pop — pop()

top len

size len

is Empty

## Stack using Array

```
class Stack:
    def __init__(self):
        self.__data = []
    def push(self, item):
        self.__data.append(item)
    def pop(self):
        if self.isEmpty():
            print("Empty Stack")
            return
        return self.__data.pop()
    def top(self):
        if self.isEmpty():
            print("Empty Stack")
            return
        return self.__data[len(self.__data) - 1]
    def size(self):
        return len(self.__data)
    def isEmpty(self):
        return self.size()==0
```

```
from StackUsingArray import Stack

s = Stack()

s.push(12)
s.push(13)
s.push(15)

while s.isEmpty() is false:
        print(s.pop())
```